University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2023
EEE-1103: Electrical Circuits

Total Marks: 70 Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) On which factors the resistance of a material depends? Relate those factors to find the resistance of a material. 3
   b) Why ammeter is connected in series and voltmeter is connected in parallel? Explain. 4
   c) Determine the unknown resistances in **Fig. 1.1** given the fact that $R_2 = 5R_1$ and $R_3 = 0.5R_1$ 4
   d) For the network in **Fig. 1.2** determine the resistance "R" : 3

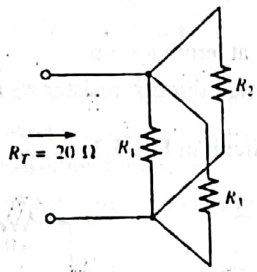

Fig. 1.1

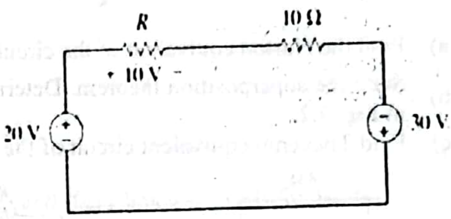

Fig. 1.2

2. a) State Current divider rule with an example. 3
   b) "The total resistance of parallel resistors will always drop as new resistors are added in parallel, irrespective of their value"- explain. 3
   c) Given the information provided in **Fig. 2.1**, find the unknown quantities: **E, R₁, and I₃**. 4
   d) Determine currents I_4 and I_5 and voltage V_2 in **Fig. 2.2**. 4

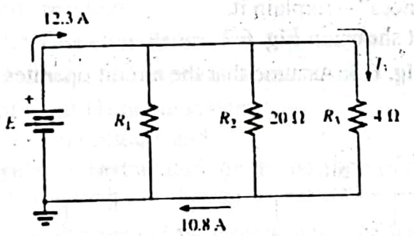

Fig. 2.1

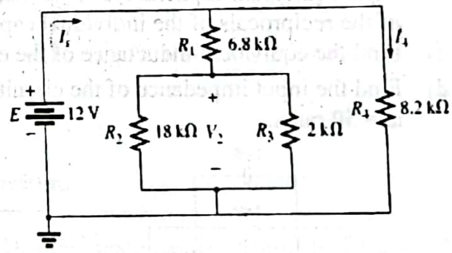

Fig. 2.2

3. a) Explain source conversion technique with an example. 4
   b) What do you mean by open and short circuits? Explain with an example. 3
   c) Find R_{ab} for the circuit in **Fig. 3.1**. 3
   d) Using a Δ-Y or Y-Δ conversion, find the current I in **Fig. 3.2**. 4

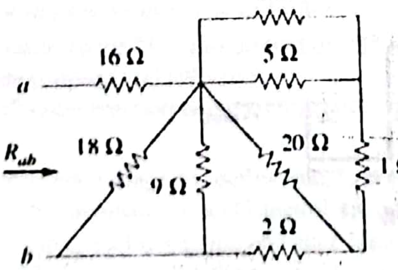

Fig. 3.1

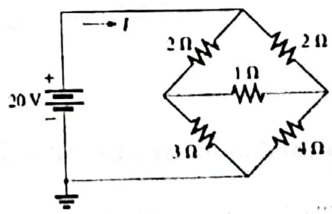

Fig. 3.2

4. a) Write the Mesh Current equations for the circuit in **Fig. 4.1.** 4

b) Find the node voltages for the circuit in **Fig. 4.2.** 4

c) i. For the network in **Fig. 4.3**, determine the value of R for maximum power to R. 6

ii. Determine the maximum power to R.

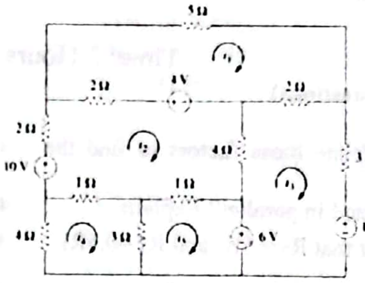

Fig. 4.1

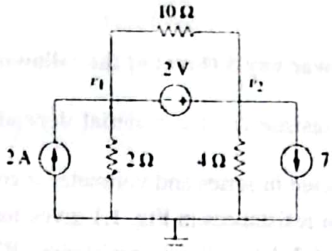

Fig. 4.2

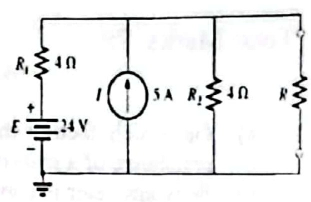

Fig. 4.3

5. a) Find the Norton equivalent of the circuit in **Fig. 5.1** at terminals a-b. 4

b) State the superposition theorem. Determine the current through resistor R_2 for the network in **Fig. 5.2.** 4

c) Find Thevenin equivalent circuit of the following circuit in **Fig. 5.3.** 6

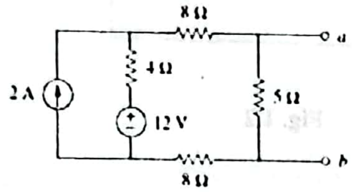

Fig. 5.1

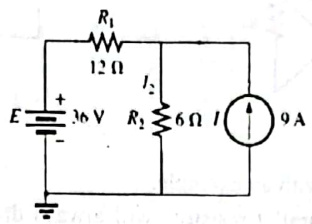

Fig. 5.2

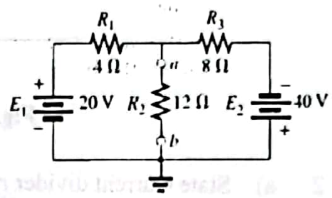

Fig. 5.3

6. a) Obtain the energy stored in each capacitor in **Fig. 6.1** under dc conditions. 3

b) "The equivalent capacitance of 'N' series-connected capacitors is the reciprocal of the sum of the reciprocals of the individual capacitances" – Explain it. 4

c) Find the equivalent inductance of the circuit shown in **Fig. 6.2.** 3

d) Find the input impedance of the circuit in **Fig. 6.3.** Assume that the circuit operates at $\omega = 50$ rad/s. 4

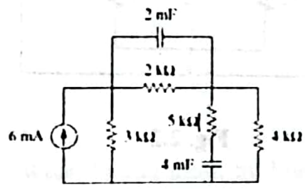

Fig. 6.1

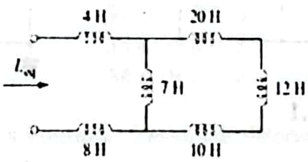

Fig. 6.2

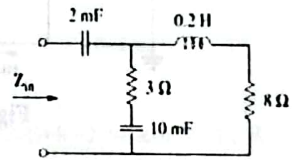

Fig. 6.3

7. a) State and explain Faraday's law of Electromagnetic Induction. 4

b) Determine the average and rms values of the square wave in **Fig. 7.1.** 4

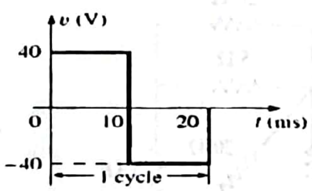

Fig. 7.1

c) Prove that average power or power dissipated by the ideal inductor is zero watts. 4

d) What do you mean by admittance and susceptance? 2